

mmTAI: Biometric-assisted Multi-person Tracking with mmWave Radar

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清華大學
Tsinghua University

Device-free Multi-person Tracking

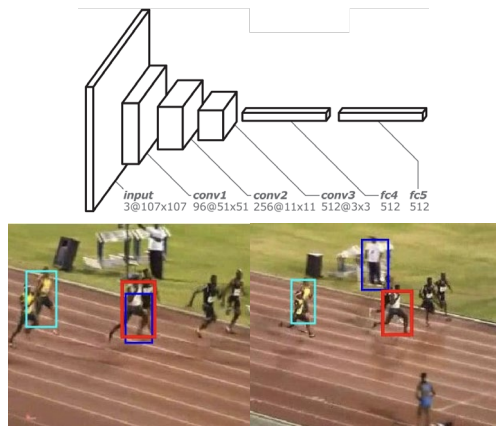


Benefits: Tracking is crucial in smart homes and elderly care scenarios.

Related Work

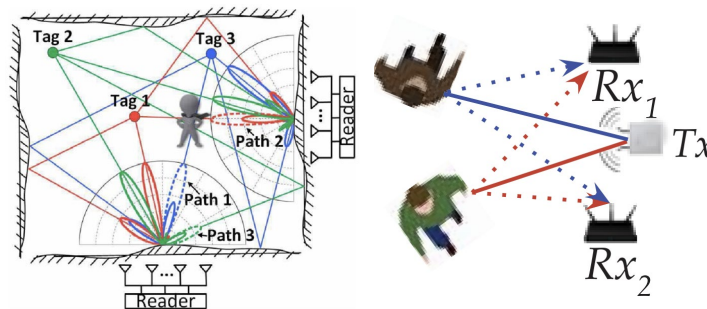


Vision-based



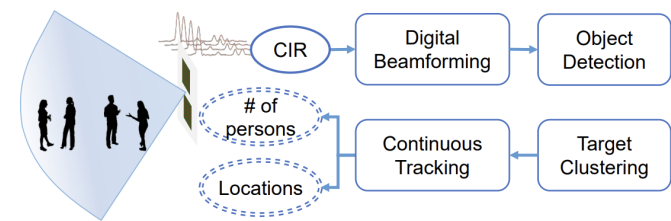
H.Nam et al., CVPR 2016

RF-based

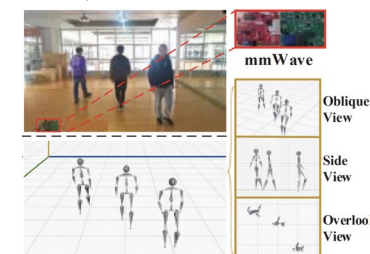


D-Watch, TON 2017
WiPolar, IMWUT 2020

mmWave-based

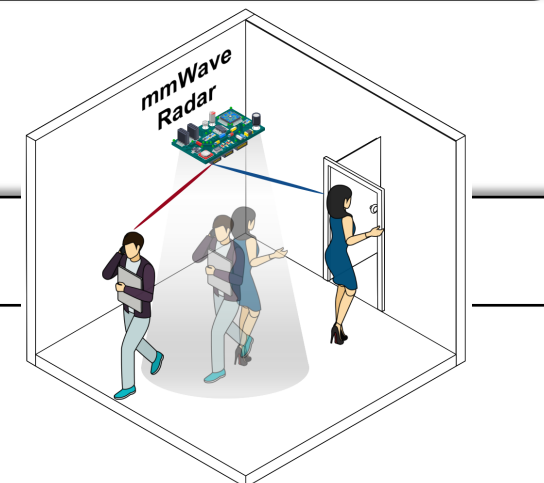


mmTrack, INFOCOM 2020



m³Track, MobiSys 2022

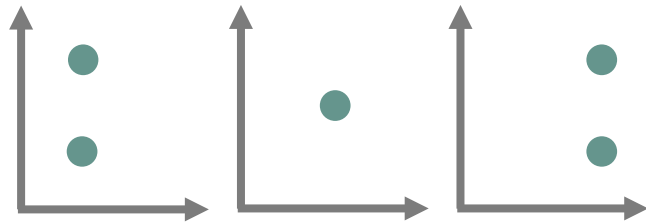
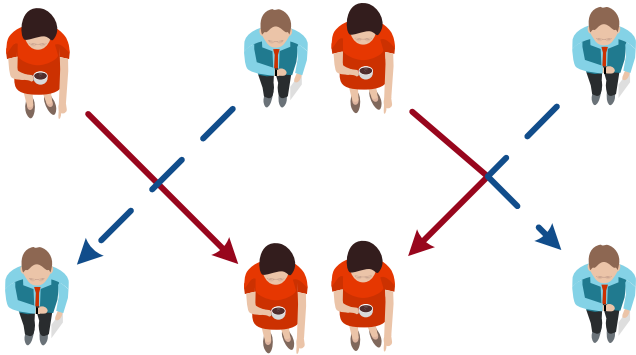
Problem: Cannot resolve trajectory ambiguity.



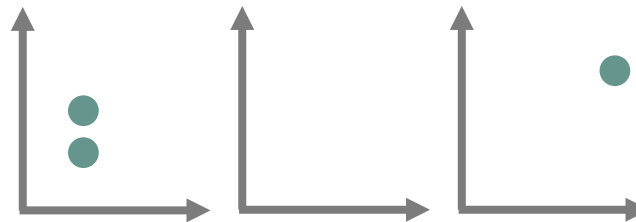
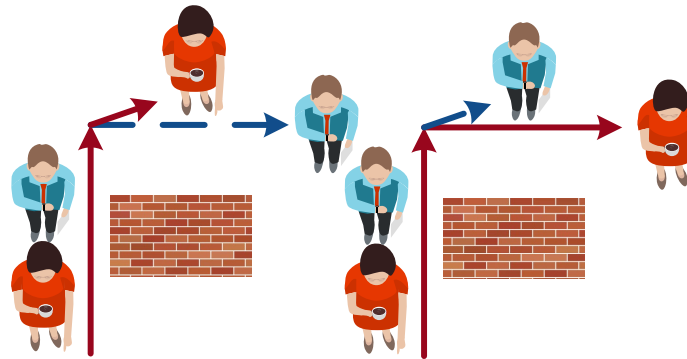
Ambiguity in Multi-person Tracking



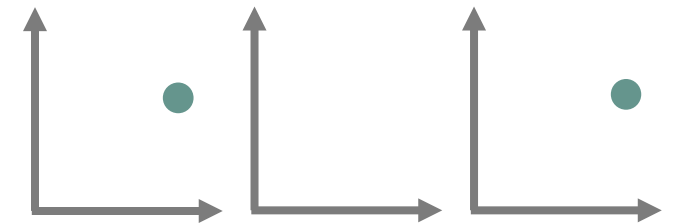
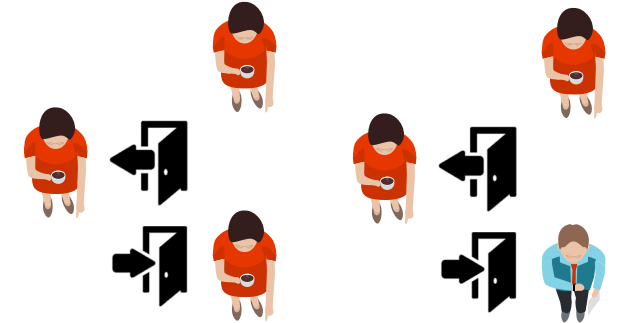
Crossover



Blockage

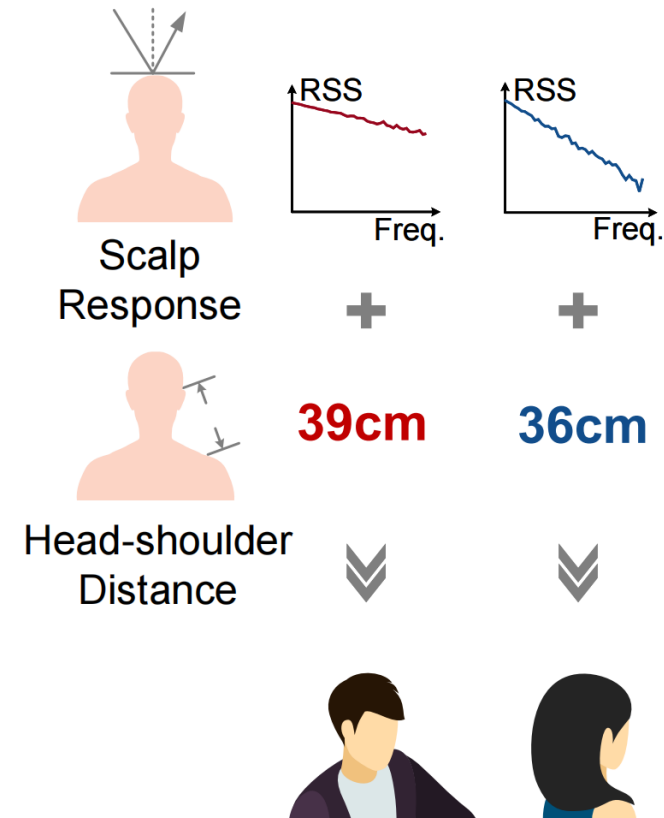
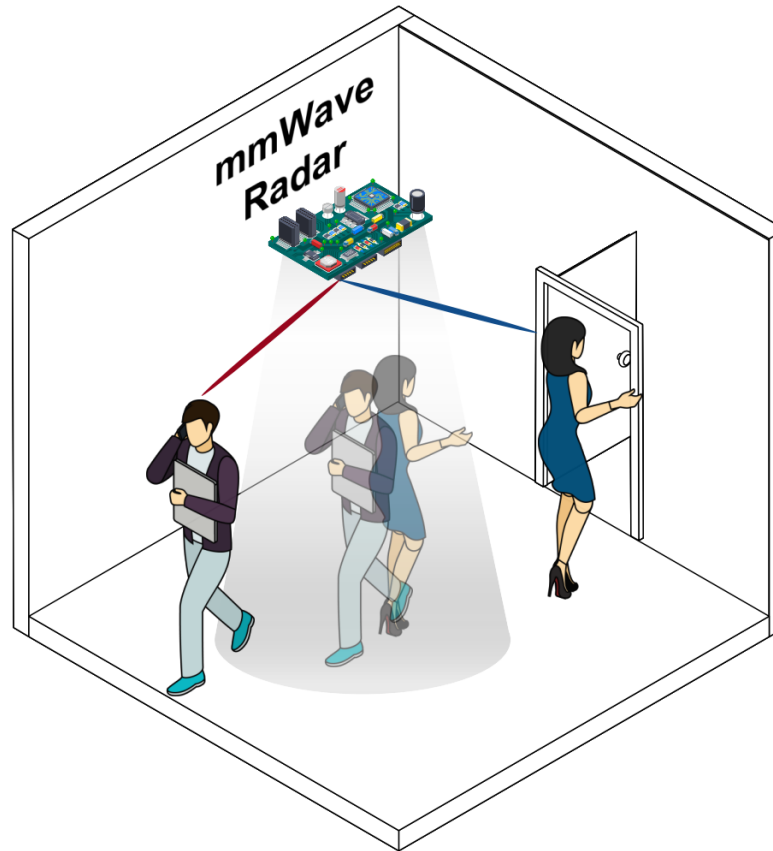


Joining/Leaving



Different trajectories lead to the same point clouds.

Our Solution



Solution: Simultaneously **differentiate** and track the persons.

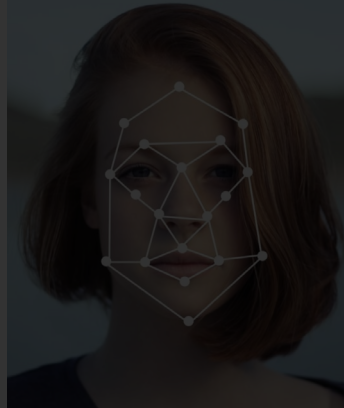
Key Insight – Biometric-assisted Differentiation



Fingerprint



Face



Gait



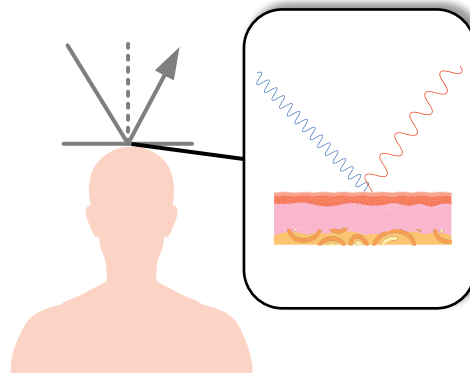
Checklist

1. **differentiating** among persons
2. **invariant** during a short period
3. **accessible** to the mmWave radar
4. **compatible** with radar resolution
5. **instantly** obtainable
6. **robust** to posture and motion

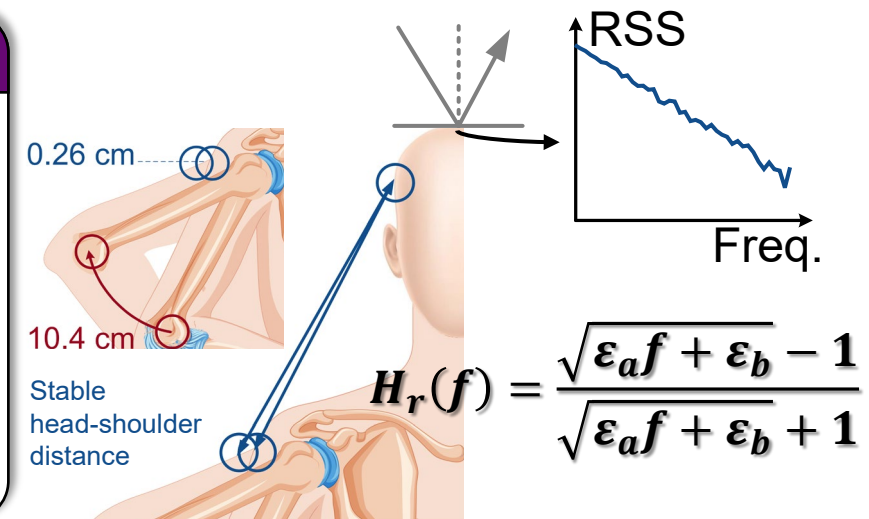
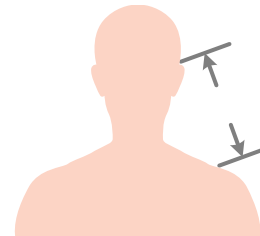
Vital Sign



Scalp Response



Head-shoulder Distance

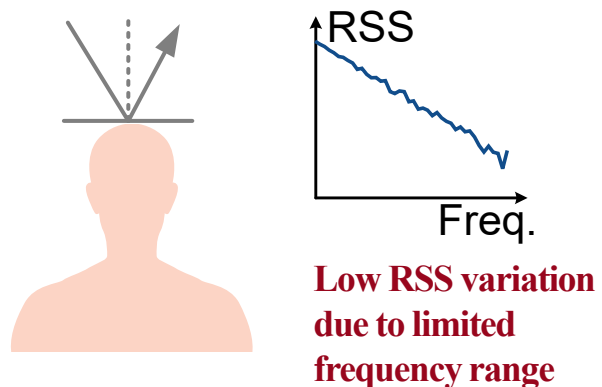
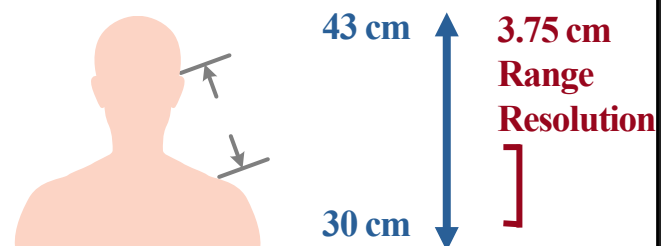


Challenges



Challenge 1

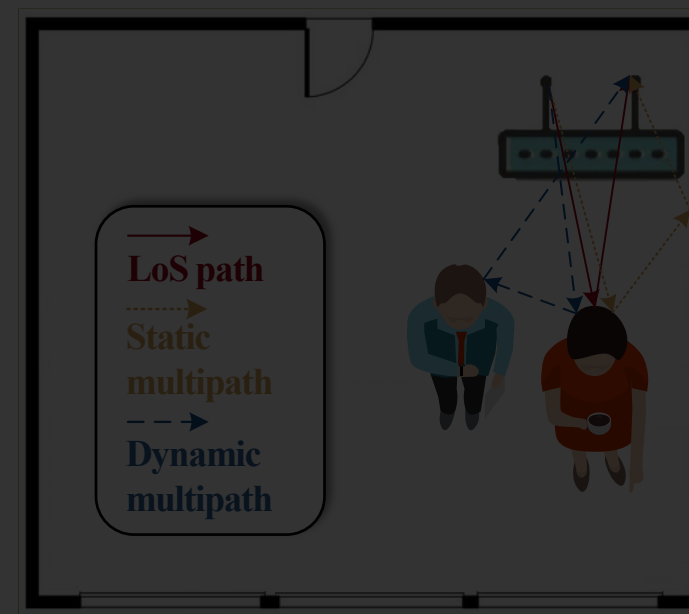
- Subtle biometric features.



The resolution of mmWave radar is limited for biometric feature measurement.

Challenge 2

- Dynamic interference on measured biometric features.

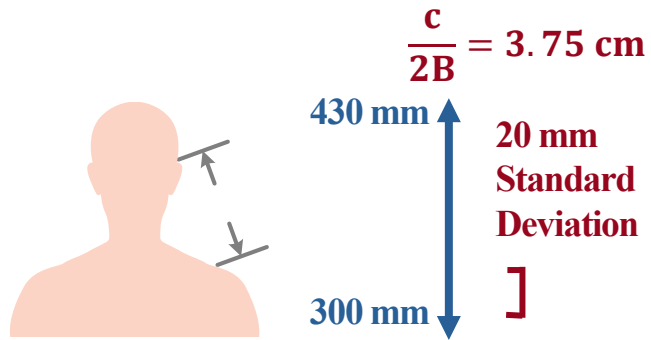


The interference results in varying biometric measurements of the same person.

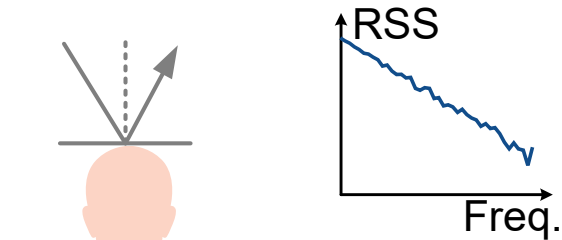
How to Extract the Subtle Biometric Features?



Head-shoulder Distance



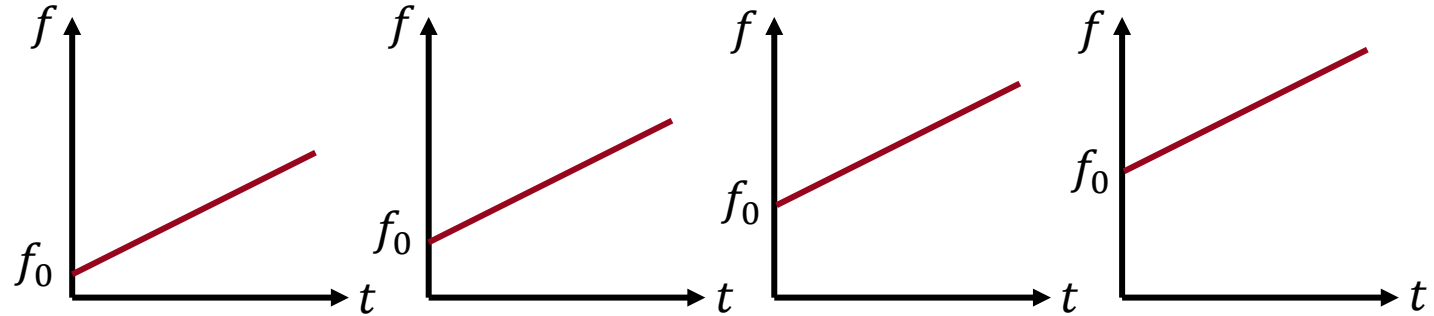
Scalp Response



Low RSS variation
due to limited
frequency range

Solution: Multiple measurements

Solution: Multi-chirp Consolidation



f_0	60.0 GHz	60.1 GHz	60.2 GHz	60.3 GHz
Phase φ	2.00	2.30	2.60	2.90

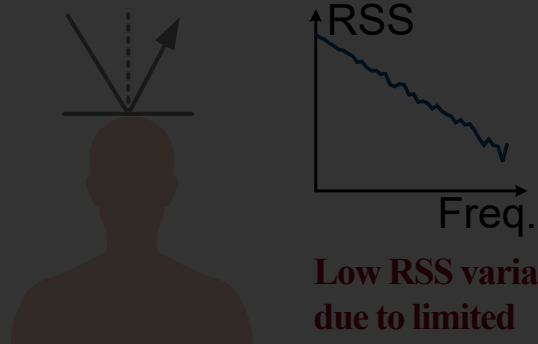
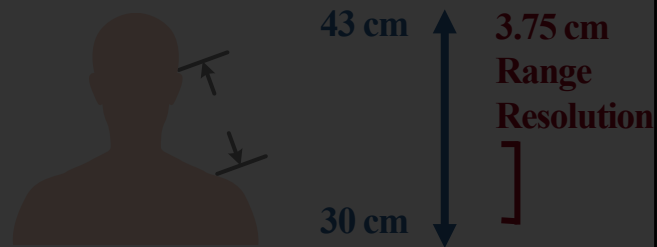
$$\varphi = \frac{1}{B} \int_{f_0}^{f_0+B} \frac{4\pi f d}{c} df \Rightarrow \frac{\partial \varphi}{\partial f} = \frac{4\pi d}{c} \Rightarrow d = \frac{c}{4\pi} \frac{\Delta \varphi}{\Delta f_0}$$

Challenges



Challenge 1

- Subtle biometric features.

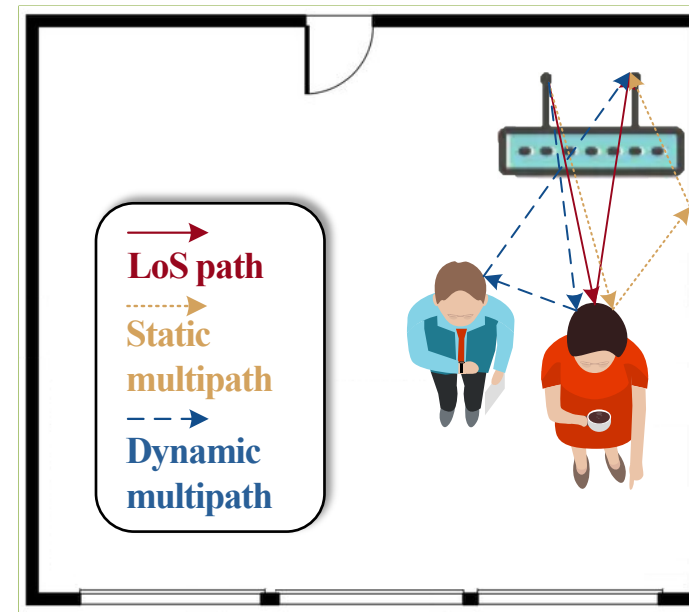


Low RSS variation
due to limited
frequency range

The resolution of mmWave radar
is limited for
biometric feature measurement.

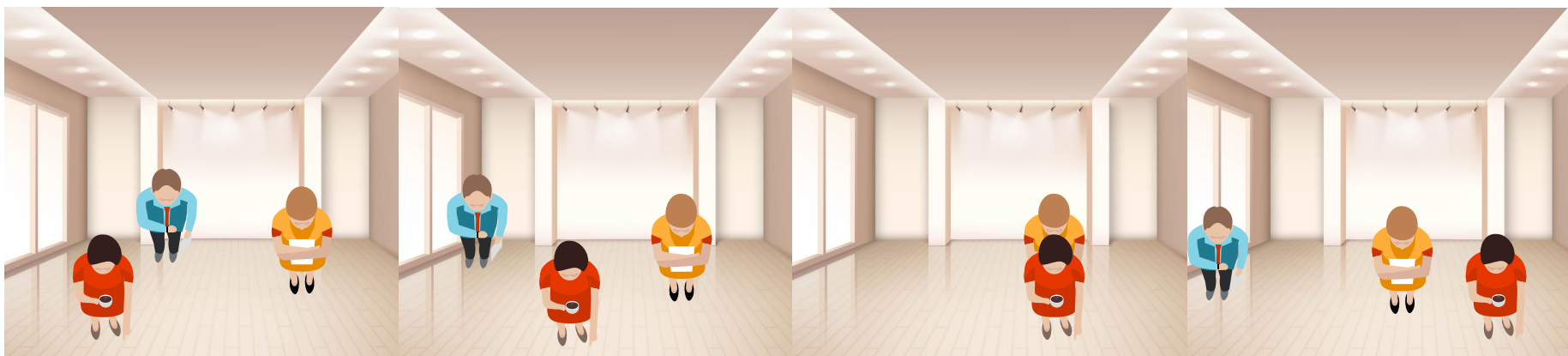
Challenge 2

- Dynamic interference on
measured biometric features.



The interference results in
varying biometric measurements
of the same person.

How to Mitigate the Dynamic Interferences?



Time

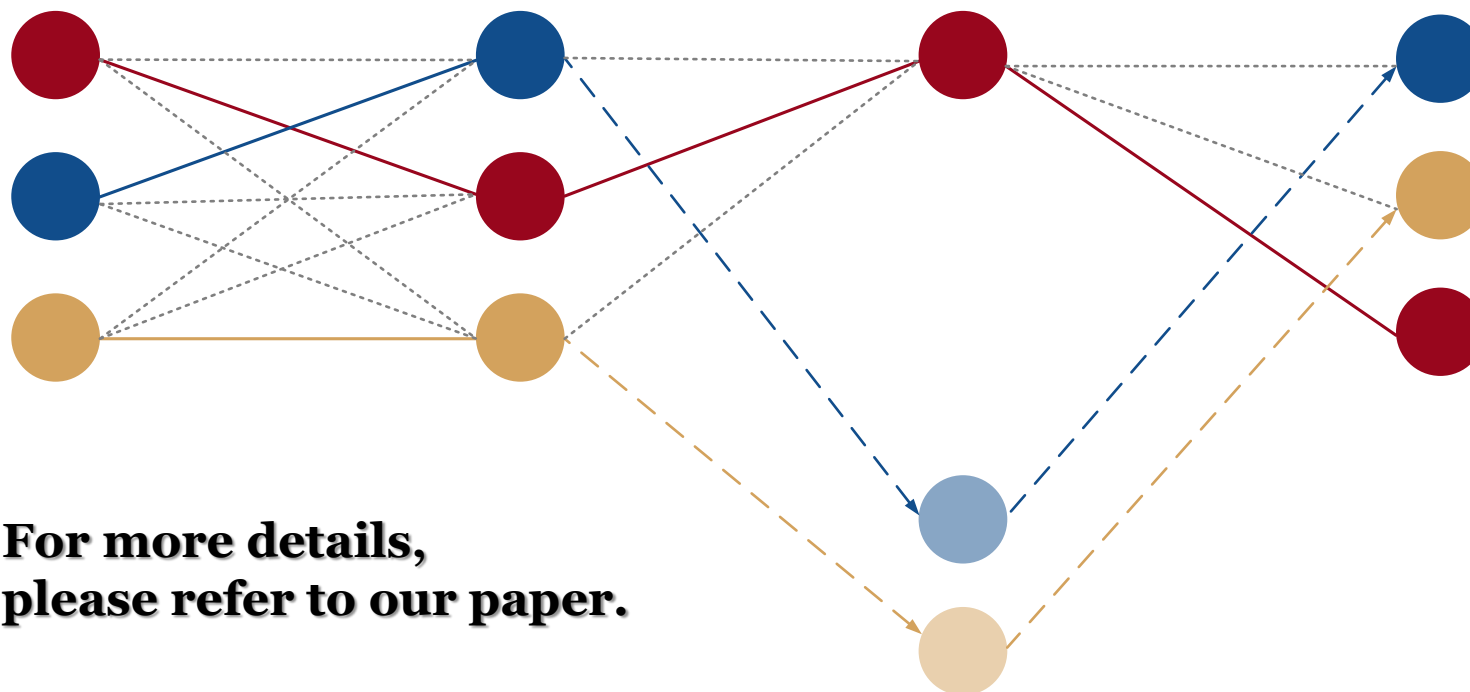
t

$t + 1$

$t + 2$

$t + 3$

Matching



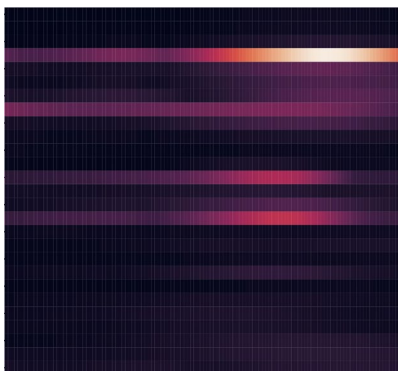
**For more details,
please refer to our paper.**

Observation:
The interferences
are invariant
within a short
period.

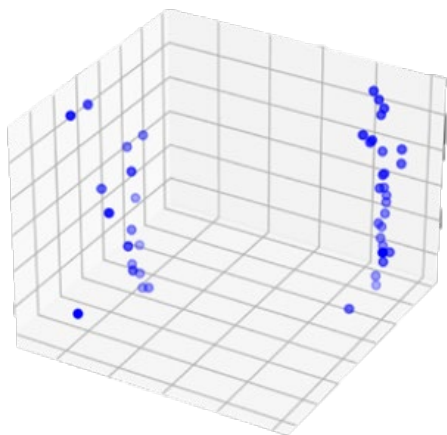
mmTAI Design Overview



Human Detection

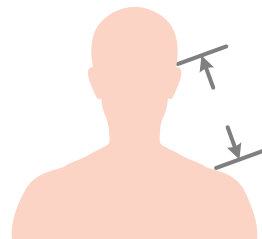


Radar Signal Preprocessing



Point Cloud Generation
and Clustering

Biometric Feature Measurement

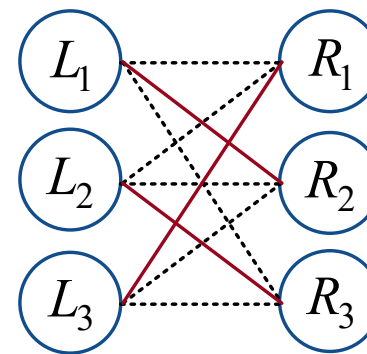


Head-shoulder Distance
Estimation

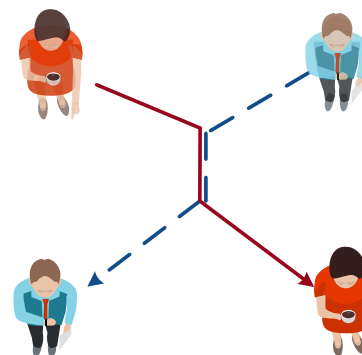


Scalp Distance Estimation

Multi-person Tracking



Short-time Person Matching

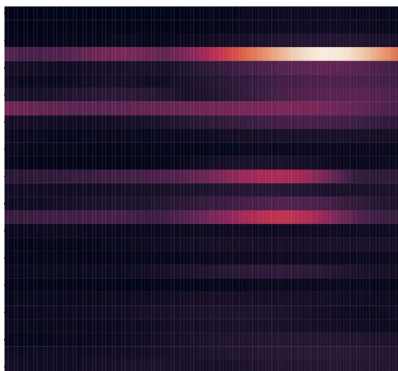


Trajectory Re-association

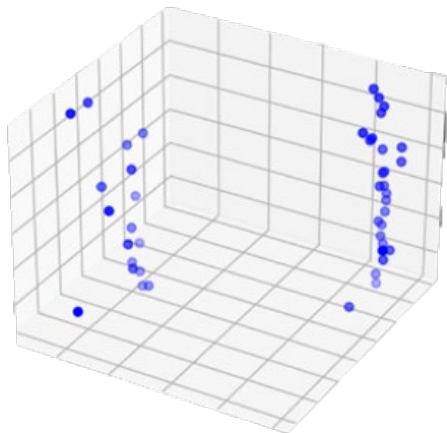
mmTAI Design Overview



Human Detection

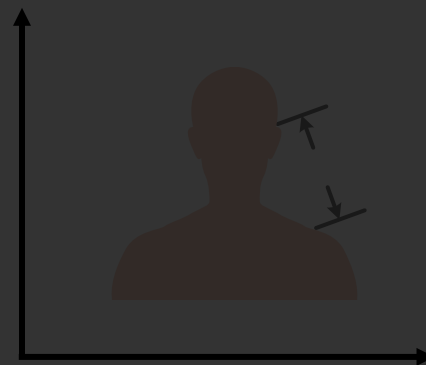


Radar Signal Preprocessing

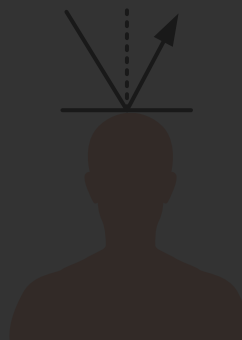


Point Cloud Generation and Clustering

Biometric Feature Measurement

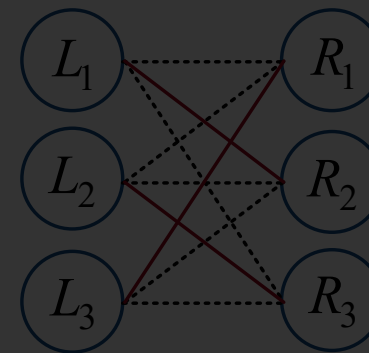


Head-shoulder Distance Estimation

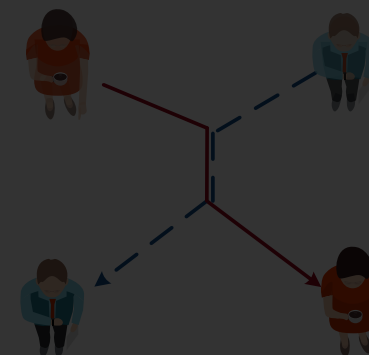


Scalp Distance Estimation

Multi-person Tracking



Short-time Person Matching

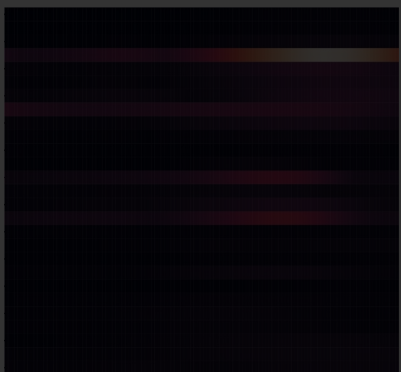


Trajectory Re-association

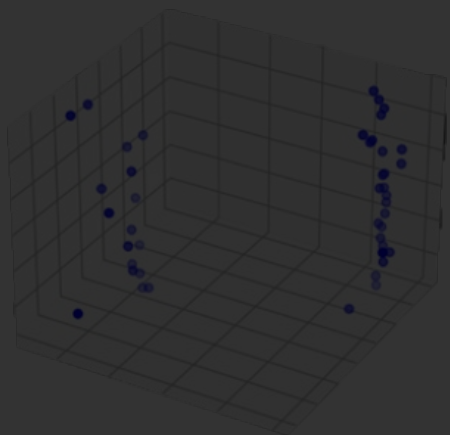
mmTAI Design Overview



Human Detection

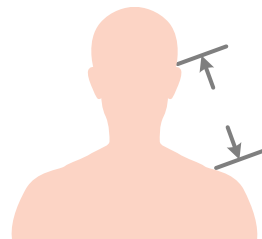


Radar Signal Preprocessing

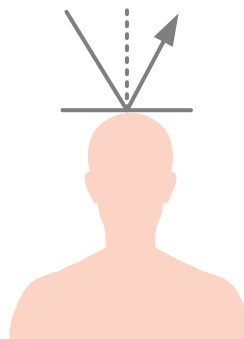


Point Cloud Generation
and Clustering

Biometric Feature Measurement

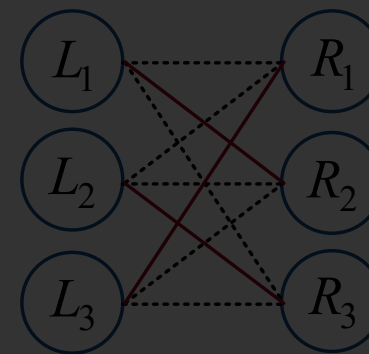


Head-shoulder Distance
Estimation

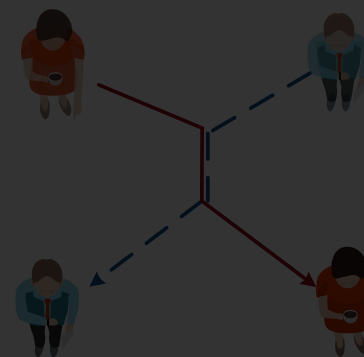


Scalp Distance Estimation

Multi-person Tracking



Short-time Person Matching

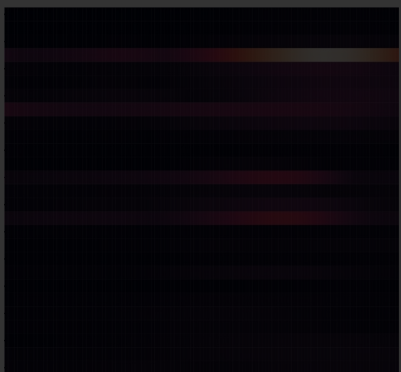


Trajectory Re-association

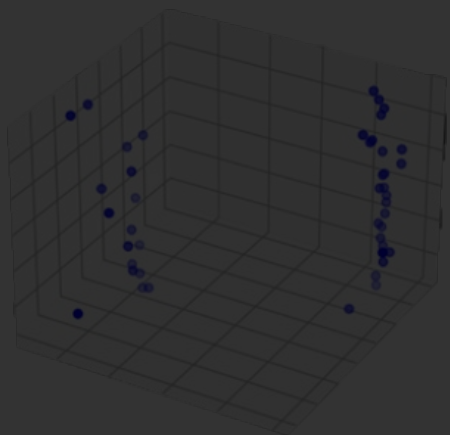
mmTAI Design Overview



Human Detection

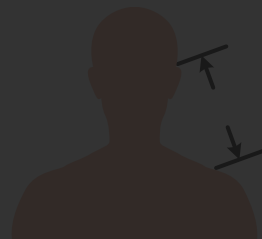


Radar Signal Preprocessing



Point Cloud Generation
and Clustering

Biometric Feature Measurement

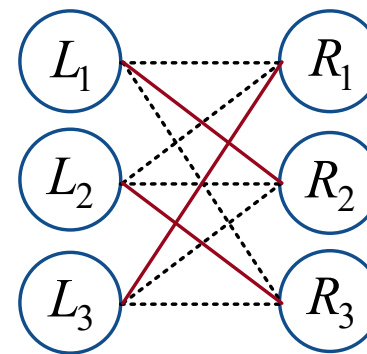


Head-shoulder Distance
Estimation

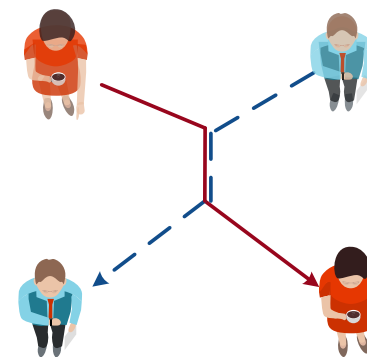


Scalp Distance Estimation

Multi-person Tracking



Short-time Person Matching

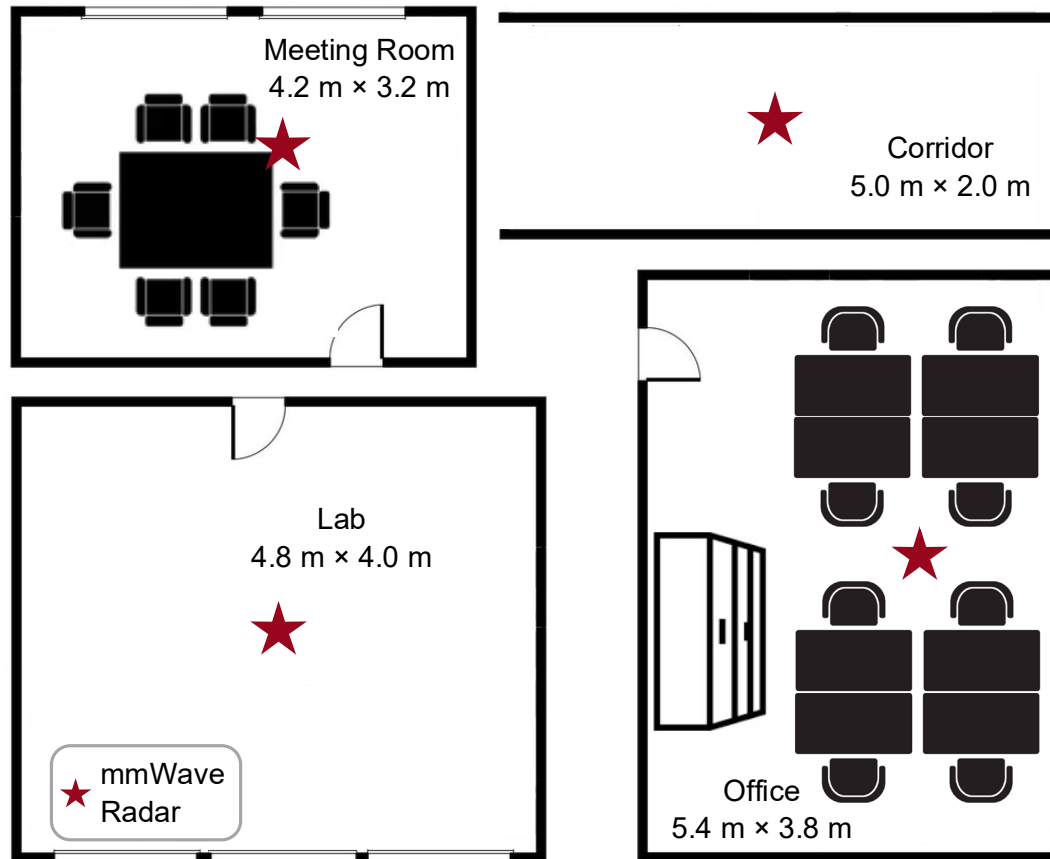


Trajectory Re-association

Implementation



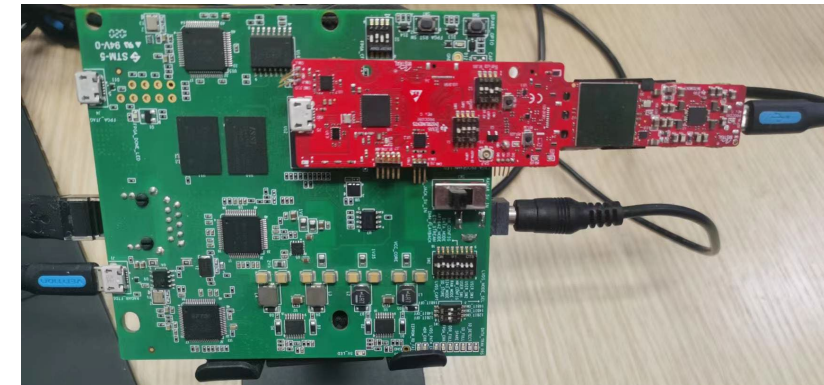
Scenarios



(The radar is placed on the ceiling, right towards the floor)

mmWave Device

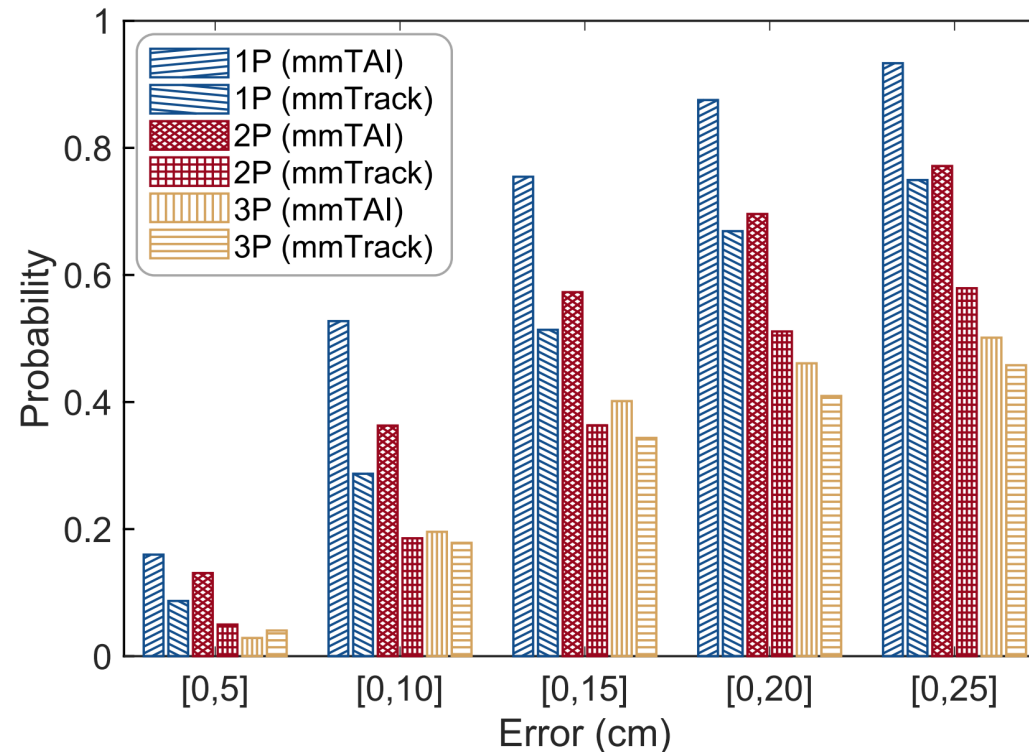
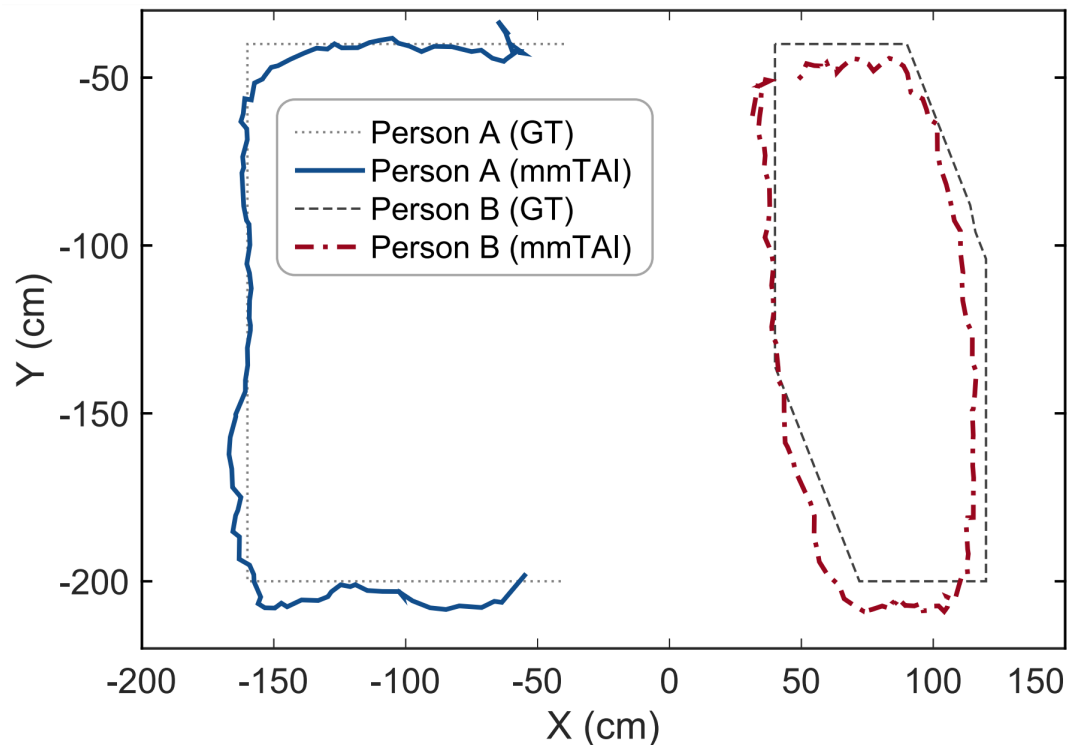
TI DCA 1000 + IWR6843AOPEVM
60 – 64 GHz, 3 TX, 4 RX
100 samples / second



Data Processing

PC (CPU: Intel i7-12700H)
Real-time processing on CPU for up to
11 persons in the tracking area

Overall Performance

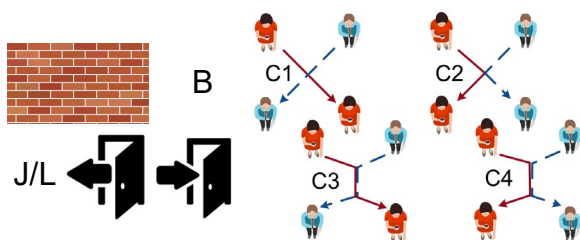
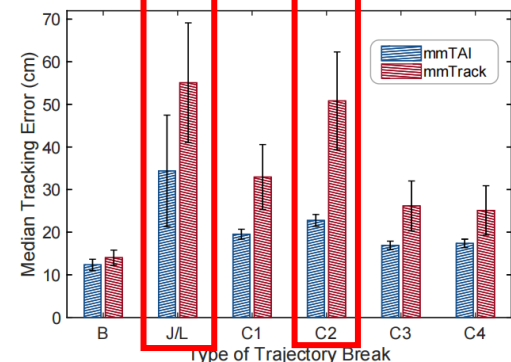
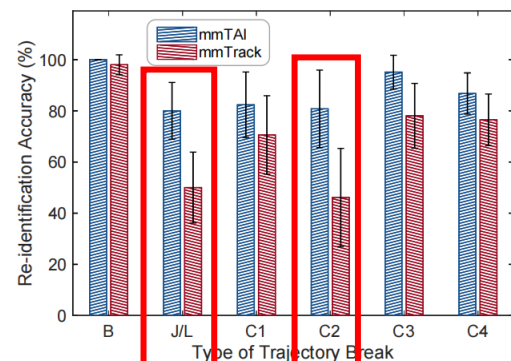


**The median error of mmTAI is 12.33cm,
35.88% less than that of mmTrack in the lab scenario.**

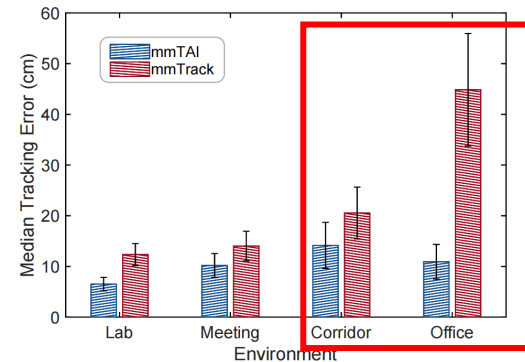
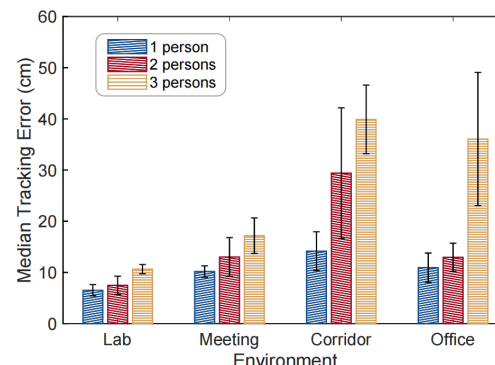
Impacting Factors



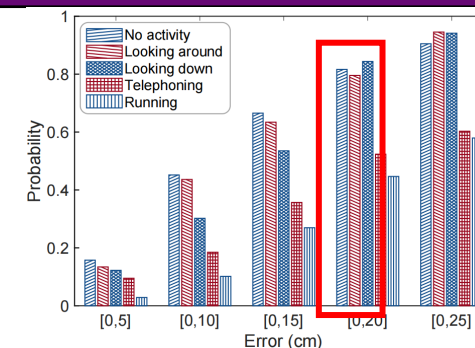
Trajectory Break



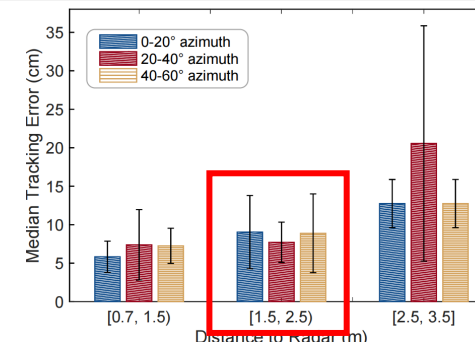
Environment



Activity



Location

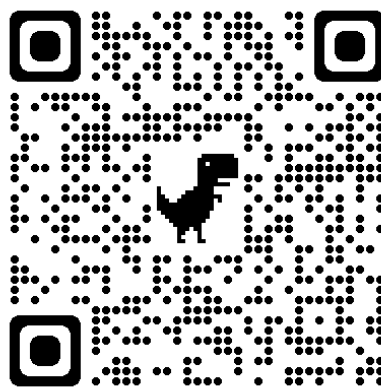


The effectiveness of mmTAI is verified under a wide range of practical conditions.

Conclusion



- mmTAI mitigates the ambiguity with biometric-assisted methods, to **enhance mmWave-based multi-person tracking**.
- The key innovation is the extraction of the **biometric features**, which is related to the **principle of RF signal reflection** and embodies the concept of **RF computing**.
- Among the multi-person tracking methods without the burden of training and running deep learning models, mmTAI achieves the **state-of-the-art localization error**.



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